INCH-POUND
MIL-L-24489C(SH)
15 April 1994
SUPERSEDING
MIL-L-24489B(SH)
21 April 1981
(See 6.11)

MILITARY SPECIFICATION

LIFEBOAT, INFLATABLE, MARK 6, 25-PERSON CAPACITY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Mark 6 inflatable lifeboat, 25-person capacity, complete, including its inflation system and survival gear, and packed within its rigid stowage container, ready for use as abandon-ship equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-S-626 - Sponges, Synthetic.
V-T-295 - Thread, Nylon.
W-B-101 - Battery, Dry.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-K-818	- Knife, Pocket, General Purpose.
MIL-W-1053	- Whistle, Ball, Plastic.
MIL-A-3339	- Anchors, Sea, Life Raft.
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-F-3747	- Flashlights: Plastic Case, Tubular (Regular, Explosion-Proof, Explosion-Proof Heat Resistant, Traffic Directing, and Inspection-Light).
MIL-T-5038	- Tape, Textile and Webbing, Textile, Reinforcing, Nylon.
MIL-D-5531	- Desalter Kits, Sea Water, Mark 2.
MIL-A-5540	- Adhesive, Polychloroprene.
MIL-B-8571	- Bag, Storage, Drinking Water.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-12647	- Pumps, Inflating-Deflating, Hand-Operated, Air, Single-Cylinder Plunger.
MIL-W-15117	- Water, Drinking, Canned, Emergency.
MIL-F-16895	- Food Packet, Survival, Abandon Ship.
MIL-R-17343	- Rope, Nylon.
MIL-C-17415	- Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.
MIL-S-17980	- Sea Marker Packet, Inflatable Survival Equipment.
MIL-M-18371	- Mirrors, Emergency Signaling, Mark 3.
MIL-C-19002	- Cloth, Coated, and Stripped, Coated Cloth-Polychloroprene on Nylon.
MIL-P-24441	- Paint, Epoxy-Polyamide General Specification for.
MIL-P-24441/1	- Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I.
MIL-P-24441/2	- Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151, Type I.
MIL-I-24490	- Inflation System Equipment, Compressed Air, for Mark 5, Mod 2 (15 Person) and Mark 6 (25-Person) Inflatable Lifeboats.
MIL-C-24491	- Container, Fiberglass Reinforced Plastic, for Inflatable Lifeboats, Mark 5, Mod 2, and Mark 6.
MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.
FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MS26529	- Oar, Metal, Sectional.
MS27826	- Plug Assembly (Sealing Clamp) Life Raft Repair.
MS27980	- Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type).

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS S8225(5)-1170434 - Inflatable Boat Equipment, Rescue Line, Detail and Arrangement.

NAVSHIPS 803-4382177 - 25 Person Inflatable Lifeboat Mark 6.

(Application for copies should be addressed to: Director, Puget Sound Naval Shipyard, DET Boston Code 244.5, 495 Summer St, Boston, MA 02210.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0902-LP-037-7010 - Instruction for Packaging, Repair and Survival Gear.

S9008-BZ-INS-010 - Instructions for Operation and Use of the Navy Inflatable Life Raft.

(Application for copies should be addressed to the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations, Title 46, 160.061 - Fishing Tackle Kit, Emergency for Merchant Vessels.

Code of Federal Regulations, Title 49, 171-178 - Hazardous Material Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. Material shall be as specified herein (see 6.6).

3.2.1 Coated cloth. Coated cloth used in the construction of the lifeboat shall conform to MIL-C-17415 except that air retention test conforming to MIL-C-19002 shall be used in lieu of Hydrogen Permeability Test Method 5460 and shall be of the following types and classes:

Part	Type	Class
Bottom	16	A
Bulkheads	16	A
Canopy	2	A or B
Canopy bows	16	A
Air cylinder holders	16	A
Cross tube	16	A
"Dee" ring patches	16	A
Equipment container	16/2	A/B
Inflatable floor	11	--
Life line patches	16	A
Lower tube	16	A
Pockets	16	A
Tape	1/ 8	A or B
Upper tube	16	A
Oar bag	16	A
Boarding ramp	16	A
Ballast bags	16	A

1/ An alternate tape fabric made of type 2 cloth of MIL-C-17415 coated on each side with 9 ounces (oz) of synthetic coating may be used.

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3.2.2 Color. Lifeboat shall be the natural color of the finished coated cloth. No pigmentation of the coated cloth is required except that the side of the canopy visible from the air when in use shall be pigmented to a color of international orange, color 12197, 22190, 28915, or 38902 of FED-STD-595.

3.2.3 Fasteners. Fasteners shall conform to MS27980.

3.2.4 Finish. Aluminum alloy parts shall have an anodic coating conforming to MIL-A-8625, type II, class 1, overcoated with two coats of epoxy-polyamide paint conforming to MIL-P-24441, one coat conforming to MIL-P-24441/1, formula 150 to a minimum of 2 mils dry film thickness (dft), and one finish coat conforming to MIL-P-24441/2, formula 151 to a minimum of 2 mils dft. Individual coats of paint shall be not greater than 4 mils. Corrosion-resistant steel and plated parts shall be left unpainted.

3.2.5 Rope. Rope, except as otherwise specified (see 6.6), shall be nylon, conforming to MIL-R-17343, and shall be as follows:

Part	Circumference (inches)
Life lines	3/4
Righting lines	1-1/8

The ends of nylon lines shall be smoothly fused or seized. When seizing is required, it shall consist of a minimum of six turns.

3.2.6 Sea painter. Sea painter shall be 5/32-inch braided nylon conforming to MIL-C-24491.

3.2.7 Thread. Sewing thread shall conform to V-T-295, type II, class 2, subclass A, size E.

3.2.8 Metal parts. Metal parts other than those specified on the drawings and herein shall be corrosion-resistant steel conforming to ASTM A 167, types 304, 316, 316L.

3.2.9 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Component parts. The completed lifeboat shall consist of a container and components conforming to MIL-C-24491, the lifeboat as shown on Drawing 803-4382177, and all equipment specified herein.

3.4 Weight. Total weight of the completed lifeboat (see 3.3) shall be not greater than 500 pounds.

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3.5 Manufacture and assembly.

3.5.1 Dimensions and shape. Dimensions and shape of the lifeboat shall be as shown on Drawing 803-4382177. For reference purposes, the bow of the lifeboat is defined as the end opposite the boarding ramp; the stern is defined as the end of the raft containing the boarding ramp.

3.5.2 Laps and seams. Laps and seams shall be secured only by vulcanizing. Sewing shall not be used in any seams or through the coated cloth of any air chamber. Sewing will be permitted only in the construction of patches, flaps, life line supports, cylinder holders, and pockets, but not in their final attachment to the lifeboat. Vulcanizing in a steam or air pressure autoclave shall be required, except for the canopy installation which may be assembled and attached by the use of room temperature curing cement conforming to MIL-A-5540, class 2. The seam overlap shall withstand a pull test load in excess of the strength of the coated fabric joined after vulcanizing. In no case shall the overlap seam be less than 3/4 inch. When subjected to a tensile load of 40 pounds per inch of width at a temperature of 200 ± 2 degrees Fahrenheit (°F), the untaped seams shall show no sign of slippage after 24 hours. Cement shall be applied in a line parallel to the edges of the seams, seam tapes, gum stripping, attachment patches, and so forth, and shall extend not less than 1/8 inch and not greater than 1/2 inch beyond the edges. Seams shall be covered inside and outside with tape conforming to MIL-C-17415, type 8, class B, not less than 1-5/8 inches wide after application, or gum stripping, having a gauge thickness of 0.032 ± 0.005 inch and a minimum width of 3/4 inch after application. The tape or gum stripping may be omitted on the outside of a lap where another layer of fabric covers the lap. The tape or gum stripping shall be applied without excessive tension, and shall be so laid that approximately an equal amount is on each side of the seam edge which it covers. Gum or fabric discs shall be used at the intersections of seam laps.

3.5.3 Bulkhead. Upper and lower tubes and the center (midship) canopy bow shall be fitted with internal vertical bulkheads, dividing each tube into two separate compartments. These bulkheads shall be of hemispherical, conical, or flexible vertical design. Bulkheads shall be located approximately as shown on Drawing 803-4382177 in order that the volume of the compartments shall be approximately equal. Bulkheads shall be of coated cloth and shall be as shown on Drawing 803-4382177 and shall be flanged for direct attachment to the inside of the tube. Flange attachment shall be reinforced with a tape not less than 2 inches wide, centerfolded into a "V", attached 1 inch on the tube at the bulkhead side opposite the flange attachment. Bulkhead attachment shall withstand a pressure of 3.8 pounds per square inch (lb/in^2), with the adjacent compartment uninflated and vented. There shall be no failure of the bulkhead or its attachment (see 4.8.3).

3.5.4 Upper and lower tubes. The upper and lower hull tubes, separate air-tight chambers, shall be constructed and assembled, one atop the other, as shown on Drawing 803-4382177. The attachment shall be reinforced inboard and outboard with two plies of cloth tape conforming to MIL-C-17415, type 8, class B, not less than 3 inches wide, centerfolded into a "V" and inserted into the crotch of the interface along the contact surface of the two tubes, approximately 1.5 inches, on

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each tube surface. Each tube shall withstand an internal pressure of 5 lb/in^2. Inner fabric reinforcement shall be provided in way of the inflation manifolds to prevent the automatic inflation system from damaging hull tubes at initial inflation. Attachment loops of grommeted tabs shall be fitted to the upper and lower tubes for securing the radar screen as shown on Drawing 803-4382177.

3.5.5 Bottom. A bottom, applied without tension, shall be attached to the lower tube and to the cross tube as shown on Drawing 803-4382177. The attachment shall be reinforced by two plies of cloth tape conforming to MIL-C-17415, type 8, class B, not less than 3 inches wide, centerfolded into a "V" and inserted into the crotch of the interface along the inside contact surface of the lower tube and the bottom, approximately 1-1/2 inches on each, around the entire peripheral distance. Lifeline patches shall be fitted to the bottom as shown on Drawing 803-4382177 for securing the survival gear bag.

3.5.6 Ballast bags. Four ballast bags shall be fitted to the bottom of each boat, as shown on Drawing 803-4382177, to improve stability.

3.5.7 Inflatable floor. An inflatable floor, consisting of two sections, shall be constructed of coated cloth conforming to MIL-C-17415, type 11 as shown on Drawing 803-4382177. Floor construction shall withstand an internal working pressure of 2 lb/in^2 and the overpressure test as specified in 4.8.2.2.

3.5.8 Cross tube. A cross tube which inflates simultaneously with the lower flotation tube shall be installed as shown on Drawing 803-4382177. It shall be located so that the lower surface of the tube is tangent to that of the hull tube. The bottom shall be attached to the cross tube (see 3.5.5).

3.5.9 Boarding ramp. A boarding ramp which inflates simultaneously with the lower flotation tube shall be installed as shown on Drawing 803-4382177.

3.5.9.1 Boarding net. A boarding net as shown on Drawing 803-4382177 shall be provided. Horizontal lines shall be fitted with 5/8-inch outside diameter (od) neoprene tubing to facilitate grasping. Tubing shall have a Shore durometer (type A) hardness of 70 to 75.

3.5.10 Canopy bows. Three canopy bows shall be provided as shown on Drawing 803-4382177. The bows for the canopy support shall be inflated simultaneously with the upper tube. The center (midship) canopy bow shall be fitted with a bulkhead (see 3.5.3 and Drawing 803-4382177).

3.5.11 Canopy. The canopy shall be as shown on Drawing 803-4382177. The exterior layer shall be cemented to the outside tangent surface of the bows, and the interior layer cemented to the inside surface of the bows. A sufficient number of 3/8-inch diameter holes shall be provided in the interior layer to allow the escape of air in order to facilitate packing of the boat. Inside and outside canopy seams shall be covered with a 1-5/8 inch wide strip of canopy fabric (see 3.2.1 and 3.2.2). The inner and outer canopy closures shall be secured with snap fasteners around the top and sides, as shown on Drawing 803-4382177, to secure them in the closed position. The bottom of the closure flaps shall be cemented to the upper (gunwale) tube. Grommeted tie patches shall be cemented to the gunwale

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tube at the openings, both inside and outside, to allow the closures to be tied, furled in the open position, with 1/8-inch diameter nylon line. A 5-inch weather flap covering the upper edge of the outer closure shall be cemented to the outer canopy fabric at the closure.

3.5.11.1 Reflective tape. Reflective tape shall be 3-inches wide 3M SOLAS, grade 3150 reflective sheeting, or equal, attached using Scotchlite primer, type 2141, or equal. Corners of all tape strips shall be rounded prior to attachment to the boat fabric. The tape shall be cut into 30 strips each measuring 8 inches long by 3 inches wide, and installed 10 each, equally spaced, on the exterior of each canopy bow as shown on Drawing 803-4382177.

3.5.12 Inflation system. The inflation system shall consist of two cylinder assemblies conforming to MIL-I-24490, type II, and shall be installed as shown on Drawing 803-4382177. The cylinder valves, after final assembly with the boat, shall be isolated from the flotation tubes by a protective packing or covering of a molded form fabricated from polystyrene foam, polyurethane foam, or polyethylene foam to prevent damage to the boat from drop impact. Rags or waste are not acceptable substitutes.

3.5.13 Stowage pockets. Stowage pockets shall be provided for the following items:

Items stowed in pocket	Number of pockets required
Hand pumps, with valve adapters	2
Sealing clamps and operation manual	1
Floatable knife	1
Rescue line	1
Spare sea anchor	1

3.5.13.1 Pockets. Pockets shall be clearly and indelibly labeled to indicate the contents thereof. Pockets shall be fitted with fasteners (see 3.2.3) and attached by vulcanizing as high as possible to allow headroom for the occupants.

3.5.14 Cylinder holders. Two cylinder holders shall be provided and installed as shown on Drawing 803-4382177. Cylinder holders shall be fitted with nylon draw-strings and closed around the bottom of the cylinder. Cylinder holders shall be of such size as to prevent contact by the cylinder or valve with the upper or lower tubes in the inflated or deflated position. Chafing patches shall be fitted on both hull tubes adjacent to the cylinder valve locations. Cylinder holders shall be sufficiently rugged to withstand the forces developed while the cylinders are being discharged and when the lifeboat is dropped from a height of 65 feet into the water. Cylinder holders shall be so located that when the cylinders are in place, the hose between each cylinder and manifold shall not be bent to a radius less than 4 inches (inside) when the lifeboat is in either the inflated or deflated condition.

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3.5.15 Lines.

3.5.15.1 Life lines. Life lines (see 3.2.5) shall be fitted around the inner and outer periphery of the lifeboat as shown on Drawing 803-4382177. The slack in the bights for the outer line shall be such that the line will extend to between 3 inches above and even with the underside of the tube. The slack in the bights for the inside line shall be not less than 2 inches.

3.5.15.2 Sea painter. Sea painter shall be coiled in the bung plug assembly in such a manner as to allow free payout of the line. Cylinder actuating cables shall be attached to the sea painter in such a manner that the cylinder valves are opened before any strain is taken against the connection to the lifeboat (see 3.7(aa)).

3.5.15.3 Rescue line. A 3/4-inch circumference highly visable polypropylene rescue line shall be provided as shown on Drawing S8225(5)-1170434 and fitted as shown on Drawing 803-4382177.

3.5.15.4 Righting line. Three righting lines (see 3.2.5) shall be installed as shown on Drawing 803-4382177. The lines shall be knotted at 1-foot intervals to facilitate grasping. The length of the righting lines, after knotting, shall be approximately 15 feet.

3.5.16 Rain catcher tubes. Rain catcher tubes shall be as shown on Drawing 803-4382177. The tubes shall be rolled and secured in the stowed condition prior to packing the lifeboat.

3.5.17 "Dee" rings. "Dee" rings shall be provided and fitted as shown on Drawing 803-4382177.

3.5.18 "Dee" ring patches. "Dee" ring patches shall be constructed and fitted as shown on Drawing 803-4382177.

3.5.19 Life line patches. Life line patches shall be installed around the inner and outer periphery of the lifeboat as shown on Drawing 803-4382177. Each patch shall withstand a 350-pound pull exerted in a direction perpendicular to the base of the patch.

3.5.20 Boarding light. A boarding light located atop the canopy, powered by a seawater activated battery, shall be provided. The battery shall be located on the bottom of the hull tube and configured in such a manner that the initial unfolding action of the boat will extract the battery sealing plugs. The wiring harness shall be fitted with a plug connector that is compatible with the battery terminals. The light fixture shall have a removable lens/dome to allow for bulb replacement, while remaining water tight. The battery shall be Chromalloy type L-12, or equal. Locations and installation of the lights and batteries shall be as shown on Drawing 803-4382177.

3.5.21 Interior lights. Two interior lights located on the underside of the forward and midship canopy bows inside the boat shall be provided. They shall be powered by a seawater activated battery separate from the boarding light battery (see 3.5.20). The wiring harness shall be fitted with a plug connector that is

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compatible with the battery terminals. The light fixture shall have a removable lens/dome to allow for bulb replacement, while remaining water tight. Installation and locations shall be as shown on Drawing 803-4382177. The battery shall be Chromalloy type L-12, or equal, installed as described in 3.5.20.

3.5.22 Radar reflective screens. Radar reflective screens shall be made of metallized (deposited silver) nylon mesh, coated white with a waterproof coating with the interstices open. The mesh shall have a minimum of 12 threads per inch in each direction and a maximum of 22 threads per inch in each direction. Screens shall be bound with 1-inch grommeted nylon tape conforming to MIL-T-5038, type II. Grommets shall be brass, size number zero. Sewn loops may be provided in lieu of grommets. Nylon line of 1/8-inch diameter shall be used for securing the bottom of the screens to the loops provided. Loops shall be so located that the bottom edge of the screens may be drawn taut. Snap fasteners for holding the screens shall be as specified in 3.2.3. Location of the screens shall be as shown on Drawing 803-4382177.

3.5.23 Equipment. The contractor shall furnish the following items of equipment with each lifeboat:

Item	Quantity
Operation and maintenance manual NAVSEA S9008-BZ-INS-010	1
Hand pumps	2
Valve adapter	2
Sealing clamp, 3 inches, MS27826-1	2
Sealing clamp, 5 inches, MS27826-2	2
Sealing clamp, 7-1/2 inches, MS27826-3	2
Survival gear bag	1
Oars, MS26529	2
Sea anchor and line	2
Fiberglass container, lifeboat	1
Rescue line (see 3.5.15.3)	1
Floatable knife	1
Oar bag	1
First aid kit	1
Sealing band, container, 6 inches wide	1

3.5.23.1 Operation and maintenance manual. Each lifeboat shall be furnished one copy of NAVSEA S9008-BZ-INS-010 operation and maintenance manual 6 by 9 inches, printed on waterproof paper with waterproof ink. The Government will provide one camera-ready copy. The contractor shall be responsible for reprinting copies of the manual in the required quantity and format (see 6.6). The copy of the manual shall be placed in the pocket. This pocket shall be stencilled in large letters on the outside "Instruction Manual."

3.5.23.2 Hand pumps. Two topping-off hand pumps conforming to MIL-P-12647, shall be provided and installed as shown on Drawing 803-4382177.

3.5.23.3 Valve adapters. Two valve adapters fitting the topping-off valve and attached to the hand pump (see 3.5.23.2) shall be provided.

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3.5.23.4 Sealing clamps. Six sealing clamps, two each of sizes 3, 5, and 7-1/2 inches, conforming to MS27826 shall be provided. The clamps shall be packaged in two sets of three clamps each with a set made up of one clamp of each size.

3.5.23.5 Survival gear bag. One survival gear bag as shown on Drawing 803-4382177 shall be provided.

3.5.23.6 Oars. Two oars conforming to MS26529 shall be provided.

3.5.23.7 Oar bag. An oar bag as shown on Drawing 803-4382177 shall be provided.

3.5.23.8 Sea anchors. Two sea anchors conforming to MIL-A-3339, type II, size 1, shall be provided. One shall be attached to the boat as shown on Drawing 803-4382177 and the second shall be stowed in the sea anchor pocket (see 3.5.13.1).

3.5.23.9 Fiberglass container. A lifeboat container conforming to MIL-C-24491 shall be provided.

3.5.23.10 Floatable knife. A floatable knife as shown on Drawing 803-4382177 shall be provided.

3.5.23.11 First aid kit. A first aid kit containing the following shall be provided:

1 - Wire splint with 3-3/4" x 30" wire fabric	2 - Gauze bandage 2" x 6 yd
2 - Sunscreen, 4 oz	2 - Gauze bandage 4" x 6 yd
2 - PVP iodine antiseptic, 1/2 oz (or 1 1-oz. package)	2 - Triangular bandage
100 - Motion sickness tablets	12 - Burn dressings, 3" x 18"
3 - Lip balm	18 - Adhesive strips 1" x 3"
2 - Traumatic wound dressings 4" x 7"	20 - Gauze pads, 2" x 2"
2 - Bandage compress, 4"	1 - Scissors
1 - Eye dressing packet	1 - Instruction sheet
	100 - Aspirin tablets

3.5.23.11.1 GFM. The first aid kit is to be equal to the kit supplied under NSN 6450-00-168-6893. Two each of the medical kit, NSN 6545-00-168-6893, will be provided GFM to the contractor within 60 days after contract to serve as production models (see 6.6). First aid kits provided in each lifeboat must meet all content characteristics and container dimensional characteristics of the medical kits provided as GFM.

3.5.23.11.2 Shelf life. The following items shall have at least 24 months shelf life remaining at the time of delivery to the Government: iodine, motion sickness tablets, eye dressing packet, adhesive strips and aspirin.

3.5.23.12 Sealing band. A container sealing band conforming to MIL-C-24491 shall be installed as specified in 3.7(ff).

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3.5.24 Protective packaging. Protective packaging for items listed below (see 3.5.23) shall conform to NAVSEA 0902-LP-037-7010. Except for the maintenance manuals, a one-unit (25 gram) bag of desiccant conforming to MIL-D-3464, type I or II, shall be included in each package. Package size, expulsion of excess air, and closure (heat sealing) shall conform to NAVSEA 0902-LP-037-7010.

Packaged items	Number of packages
Hand pump with detached hose and adapter	2
Operation and maintenance manual	1
Sealing clamps, 3, 5, and 7-1/2 inches	6
Floatable knife	1
Sea marker, fluorescein	1
Flashlight batteries	1
Motion sickness tablets, dimenhydrinate, 50 milligram	1

After protective packaging, the items shall be placed in their respective stowages in the lifeboat.

3.5.25 Survival gear. The equipment listed in table I shall be provided and packed into the survival gear bag (see 3.5.23.5) as shown on figure 1, which shall be closed as shown on figure 2.

TABLE I. Survival gear.

Item	Specification	Quantity
Packet, food 1/ 3/	MIL-F-16895	125
Water, plastic container, 16-ounce; Commercial 50 3/	MIL-W-15117	75
Opener, beer can type	MIL-W-15117	6
Desalter kit, type II	MIL-D-5531	5
Storage bag, drinking water, size A	MIL-B-8571	2
Flashlight, type II, style 1	MIL-F-3747	1
Battery, dry, flashlight (alkaline, size D) 2/ 3/	W-B-101	4
Sea marker, fluorescein, canister type	MIL-S-17980	1
Mirror, signalling emergency, type II	MIL-M-18371	1
Sponge, cellulose, type II, approximately size 2X4.5X6.75	L-S-626	2
Knife, pocket	MIL-K-818	1
Whistle, signalling, plastic, type II	MIL-W-1053	1
Measuring cup, plastic, 8-ounce (graduated)	Commercial	2
Motion sickness tablet, dimenhydrinate, 50 milligram 3/	Commercial	250
Bailer, plastic, 2-quart capacity	Rubbermaid, part 2885, or equal	1
Kit, first aid	See 3.5.23.11	1
Kit, signalling	See 3.10	1
Fishing kit, survival	CFR 46, 160.061	1
Flashlight bulbs (PR6)	Commercial	1

See footnotes at top of next page.

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1/ Each food packet shall be provided with the instruction: "Rations for one person for 1 day" and stamped with date packed into survival gear bag.
2/ Commercial "alkaline energized" batteries may be substituted.
3/ These items shall have at least 24 months shelf life remaining at the time of delivery to the Government.

3.6 Stowing and packing of equipment on board the boat.

3.6.1 Survival gear. The packed survival gear bag (see 3.5.25) shall be secured in its place adjacent to the cross tube as shown on Drawing 803-4382177.

3.6.2 Oars. Oars shall be packed in the oar bag (see 3.5.23.7).

3.6.3 Equipment. Remaining items of equipment shall be stowed in the pockets specified in 3.5.13.1.

3.6.4 Sea painter. Sea painter shall be coiled and installed in the bung plug/painter line assembly in such a manner as to allow free payout of the line. The bung plug/painter line assembly shall be installed on the life raft container in the same manner as the grommet assembly shown on Drawing 803-4382177.

3.7 Folding instructions. Lifeboat shall be folded for packing into its container (see 3.5.23.9) as follows, care being taken in the handling to preclude damage to the boat or its inflation system components or discharging of the air cylinders. Talc/talcum used in the packaging process shall be free of asbestos and asbestiform-like materials.

(a) The working area shall be clean, dry, and free from obstructions, snags, and other objects or obstacles that may inflict damage to the boat during handling. The entire boat shall be dusted lightly with talc.
(b) The packed survival gear bag (see 3.5.25) shall be located in place (see 3.6.1) along the longitudinal centerline of the boat as shown on figure 3, step 1. The closure end of the bag in this position shall be located even with the after side of the center canopy arch tube.
(c) With the life raft laid out flat, one end of the 100-foot long (coiled) painter shall be tied to the bow end with a bowline knot (see Drawing 803-4382177 for rigging the painter). Ten feet of slack shall be allowed and a 2-inch diameter loop placed in the painter. The balance of the coiled painter shall be placed just outside of the folding area.
(d) The packed oar bag (see 3.5.23.7) shall be placed along the starboard (right) side of the survival gear bag.
(e) The oar bag shall be tied to the survival gear bag and the survival gear bag shall be secured in place.
(f) The floor sections (see 3.5.7) shall be installed, securing them in place using the tabs and lines intended for this purpose.
(g) The stern closure shall be rolled flap down, leaving the portal open, and the bow portal shall be closed and secured by its snap fasteners.

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(h) The boat shall be completely deflated, evacuating all the air from within the various air compartments until the boat is flat. The six topping-off valves (two in the inflatable floor sections and one in each hull and gunwale tube compartment) shall be closed handtight (counter-clockwise). The boat shall be rechecked to ensure the valves are closed.

(i) Both the sea anchor lines shall be attached to the bow "dee" ring, by means of a bowline knot, with the free ends seized to the standing parts. The spare sea anchor shall be inserted into its stowage pocket and the pocket shall be snapped closed.

(j) Two fully charged air cylinders conforming to MIL-I-24490 shall be secured in the cylinder holders (see 3.5.14), one on each side of the boat. The cylinder having the shorter (10-inch) pull cable shall be installed in the pocket on the starboard side.

(k) The various elbow and hose connections between the manifolds and cylinder valves shall be assembled conforming to MIL-I-24490 using the torque values specified.

(l) After step (k) above is accomplished, it shall be ensured hoses are not kinked, and that each lays in a plane parallel to the tube and bends from the inflation valve toward the cylinder bottom then back to the manifold.

(m) The painter line shall be secured to the towing line and rigged as shown on Drawing 803-4382177.

(n) The water-activated batteries (see 3.5.20 and 3.5.21) shall be inserted into their pockets on the underside of the hull tube, rigging them with the line as shown on Drawing 803-4382177 to provide for automatic removal of their plugs upon inflation of the boat. The electrical plug for each light shall be connected to its respective battery.

(o) The stern section of the boat shall be folded toward the bow, pulling it hard against the survival gear bag within the boat (see step 2 of figure 3).

(p) All of this section forward of the bag shall be folded back over the bag (see step 3 of figure 3).

(q) The boarding ramp shall be folded forward, neatly over the previous folds (see step 4 of figure 3).

(r) The boarding ladder shall be folded back onto the unfolded (forward) section of the boat, spreading it out flat (see step 5 of figure 3).

(s) The forward section of the boat shall be folded back, to the front of the survival gear bag (see step 6 of figure 3), laying the loose sea anchor on the bottom of the bow, so that the sea anchor will deploy early in the inflation process.

(t) This section shall be folded hard over the survival gear bag and previous folds, steps (o), (p), and (q) above. Following this fold, the hull tube inflation "dee" ring (see piece 20, section 8-A, Drawing 803-4382177) shall be ensured to be adjacent the port (left) cylinder valve (see step 7 of figure 3).

(u) The pull cable shall be attached from the port side cylinder valve to the "dee" ring.

(v) The port side of the boat shall be folded snugly over the survival gear bag tucking the port cylinder against the starboard side of the survival gear bag (see step 8 of figure 3).

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(w) The starboard (right) side of the boat shall be folded over hard against the survival gear bag, so that the cylinder rests on the top of the folded boat (self-explanatory, not shown on figure 3). This cylinder valve has the shorter (10-inch) pull cable.

(x) Using load binders (Gripper Model 106, or equal), the folded boat shall be compressed making sure that the binder straps are over the cylinder, step (w) above. Properly done, the folding procedure, steps (o) through (w) above shall produce a rounded package no greater than 48-inches long, 26-inches high, the maximum size possible for fit into the rigid container.

(y) With the load binders still in place, compressing the boat, the boat shall be placed into the container (see 3.5.23.9) lower half.

(z) The center of gravity of the folded boat in the container lower half shall be checked by allowing it to rock freely, side-to-side. The container half should come to rest with its gasket surface level. If not, the boat shall be shifted until it is in proper balance.

(aa) The exposed pull cable hook shall be attached to the bight in the painter line (approximately 10 feet from the bow "dee" ring attachment).

(bb) Allow 2 feet of the 100-foot painter to extend through the grommet's bung hole and into the slit in the bung plug. The bung plug/painter line assembly shall be installed in the container grommet slot.

(cc) Release and remove the load binders taking care not to shift the boat pack.

(dd) Place the upper container half on the boat with the hole aligned with the lower container half.

(ee) Attach load binders around the outside of the container with the ratchets centered on top of the container. Tighten the load binders to close the container being careful not to pinch the boat between the container halves. With the container closed, rock the boat again to check for balance.

(ff) While the load binders are still tightened on the container, install the rubber sealing band over the horizontal container joint. The band shall be equally spaced over the joint (see figure 4).

(gg) Position the brass securing bands on the container so that the weak link is located on the right side when facing the bung hole, approximately 30 degrees up from the horizontal joint (see figure 4). Cement sealing band to lower half of fiberglass container in four places. Use cement sparingly in order to avoid getting it on the upper half of the container or in the seam.

(hh) Tighten the bands until no space can be seen between the bands and the container. Secure each band with a banding buckle and locate the buckle approximately 4 inches below the container joint. Release the load binders slowly, allowing the container to expand against the banding straps. Remove the load binders.

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(ii) Lift up the rubber band seal covering the bung plug hole and remove the bung plug from the painter line. Tie an overhand knot in the painter line approximately 24 inches from the end. Reinstall the bung plug on the end of the painter line and slide it up to the knot. Reinstall the bung plug in the container and reposition the rubber sealing band. Ensure the painter line extends out over the top of the sealing band. Tape the end of the painter to the container using duct tape (see figure 4).

(jj) After the packing is complete, again check the lifeboat for proper balance by allowing the boat to rock freely from side to side. The container shall come to rest. Measure the height of the horizontal joint at the closest point to the floor. This distance shall be greater than 11 inches. In the event this does not occur, the container shall be opened and the boat repositioned until levelness is achieved.

3.8 Marking. Markings shall be legible, gothic letters and numbers, and shall be thoroughly dry prior to packing. The ink used shall have an acidity (pH) of 5.0 to 8.5 and shall contain no ingredient which is deleterious to the material being marked.

3.8.1 Instructions. Instructions relative to topping-off and deflating shall be provided on the tubes near the topping-off valves. Instructions relative to topping-off and deflating shall read as follows:

"TO INFLATE - Insert pump with adapter into valve. Open valve to right (clockwise). Pump until firm. Close valve to left (counter-clockwise)."
"TO DEFLATE - Turn valve to right (clockwise)."

3.8.2 Labeling. Each lifeboat shall have fitted on the top of the lower tube, at the stern, a molded patch, or an equivalent label (see 6.6) containing the following information:

"Lifeboat, inflatable, Mark 6
Stock No._____
Maximum capacity - 25 persons
Build for the United States Navy
Contractor, city, month, year
Contract No._____
Serial No._____"

3.8.2.1 Serial numbers. Serial numbers shall be assigned to these lifeboats by the contractor. Each lifeboat shall have a unique, systematic serial number regardless of contracts or purchase orders.

3.8.3 Righting instructions. Each lifeboat shall have righting instruction patches fabricated from canopy fabric (see 3.2.1) as specified in 3.8.3.1 and 3.8.3.2. Outside corners of all patches shall be rounded. Characters shall be 1 inch high, except that the words "RIGHTING INSTRUCTIONS" in the titles shall be 2 inches high.

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3.8.3.1 Hull tube instruction patches. Two patches (9-1/2 inches by 28 inches each) containing the following legend shall be located one on the port side and one on the starboard side, inverted, just forward of each inflation cylinder:

"RIGHTING INSTRUCTIONS

1. Three men climb up onto boat bottom.
2. Further instructions on bottom."

3.8.3.2 Bottom instruction patches. Two patches (24-1/2 inches by 28 inches) shall be located amidships one each port and starboard on the boat's bottom oriented with the downward pointing arrow of step 2 pointing outboard containing the following information shall be provided:

"RIGHTING INSTRUCTIONS

1. Each man take hold of one righting line.
2. Position feet on either side of righting line while standing over this edge of the boat.
3. Pull up on righting line till taut.
4. Fall backwards hard and together while pulling righting line hand over hand toward you."

This patch shall be fabricated from either one or two fabric pieces at the contractor's discretion.

3.8.4 Code symbols. International Morse Code for the English alphabet A through Z and for Arabic numerals zero through nine shall be printed on an 8- by 12-inch area or on a piece of canopy fabric and cemented on the inside of the boat canopy adjacent to the stern entrance. Each code symbol shall be identified by its appropriate letter or numeral counterpart next to the symbol.

3.9 Performance.

3.9.1 Inflation speed. When tested in accordance with 4.8.1, the life-boat shall inflate to its designed shape in not greater than 30 seconds. After a stabilization period, the gauge pressure in any compartment shall be not less than 2.3 lb/in^2 nor greater than 3.8 lb/in^2.

3.9.2 Over-pressure discharge. When tested in accordance with 4.8.2, the pressure after 20 minutes shall be not less than 4.6 lb/in^2.

3.10 Government-furnished equipment. The following equipment will be Government-furnished (see 6.6):

(a) Kit, signalling, NSN 2T-1370-00-309-5028.

3.11 Workmanship. Wherever repair is found to be necessary, it shall be performed by heat vulcanization either by the local application of heat or by autoclaving the entire lifeboat. A maximum of 15 repairs or reworks per lifeboat shall be permitted. A maximum of three bulkhead repairs shall be permitted to any

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one boat, while not more than 15 percent of the lifeboats, in any contract or order, may have in excess of one bulkhead repair. A multiplicity of pin holes in any particular section is not considered repairable. Both an inner and outer fabric patch shall be used to repair or rework tears or cuts. Treating or swashing the inflatable members of the lifeboat with latex or any other sealing agents is not permissible (see 4.5.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in 4.6 and 4.8.1 through 4.8.3 (see 6.3).

4.4 First article sample. The first article samples shall consist of the following as specified (see 6.3):

(a) One lifeboat complete with equipment and stowage container.
(b) Four yards of all coated cloth used (vulcanized).
(c) Four yards of all cloth used (uncoated).
(d) Ten samples of life line patches cemented and vulcanized to a 7-inch diameter piece of tube fabric as specified in 3.2.1 and conforming to MIL-C-17415.
(e) Four sample "dee" ring patches vulcanized to an 18-inch by 14-inch piece of tube fabric as specified in 3.2.1 and conforming to MIL-C-17415.

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(f) Four vulcanized seams 7 inches wide by 12 inches long of tube fabric as specified in 3.2.1 and conforming to MIL-C-17415. Seams shall be prepared and vulcanized in the same manner as the lifeboat tube seams.

4.5 Quality conformance inspection. Each lifeboat shall be inspected for all characteristics which determine whether or not the unit will perform its primary life saving function and which depend upon workmanship rather than design factors. Inspection shall consist of the following and shall be conducted in the order listed:

Test	Test method
Inflation test	4.8.1
Overpressure leakage test	4.8.2 and 4.8.2.1
Overpressure leakage test, floor	4.8.2.2
Working pressure and bulkhead test	4.8.3
Lap seam strength test	4.8.5
Examination	4.6

4.5.1 Lap seam sample. The contractor shall prepare two untaped seam samples per lifeboat and shall subject these samples to the tests specified in 4.8.5 (see 6.3). These seam samples shall be constructed of the same lot of fabric utilized in the manufacture of the lifeboat they represent. These untaped seam samples shall have a minimum of 3/4 inch lap. They shall be cured simultaneously with the specific lifeboat they represent.

4.5.2 Repair and replacement. Lifeboats not meeting the requirements of the inspection specified in 4.5 shall be corrected prior to submitting the lifeboats for acceptance by the Government (see 6.3). Treating or swashing any inflatable member of the lifeboat with latex or any other liquid sealing agents in order to pass the inspection specified in 4.5 shall not be permitted. It is the contractor's responsibility to recharge the cylinders discharged during inspection. Only cylinders, valves, hoses, fittings, and pull cables which have actually been used during the inspection shall be installed on the lifeboat with the following exception: The replacement of a faulty or leaking cylinder valve is permitted. The inflation hoses shall be replaced prior to the lifeboat being submitted for Government acceptance. This is not intended to restrict the cylinder assembly to the lifeboat on which it was tested.

4.5.3 Random sample inspection. Prior to shipment, one completed boat in every lot of ten will be randomly selected by the cognizant Government inspector and subjected to the test in 4.8.1. Subsequent to the performance of this test, both inflation hoses in the tested lifeboat shall be replaced. Failure of the container to open properly, failure of the inflation system to perform, or painter line pull force greater than 40 pounds, shall be cause for rejection of the lot. The manufacturer then has the option of inspecting each lifeboat in the rejected lot for the discrepant condition, or providing a new lot which shall be inspected as required herein.

4.6 Examination. Each completed lifeboat shall be carefully examined to determine conformance with the requirements of this specification and for such additional examination as may be considered necessary to ensure suitability for the purpose intended. As a minimum, examination shall check for defects as

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specified in table II and review of repair record(s) as part of examination process. No lifeboat will be accepted by the Government until all defects are corrected.

TABLE II. Types of defects.

Defects
Coated cloth of the inflatable compartments damaged, not properly cured, not as specified.
Poor adhesion, improper vulcanizing.
Hoses damaged. End connections not torqued as specified. End connections cross-threaded.
Actuating "dee" ring for the lower tube cylinder valve not installed as specified.
Evidence of sewing in the seams of, or through the cloth of air chambers.
Valve pull-cables damaged, or not connected as specified.
Topping-off valves not fully closed as specified.
Evidence of treating or swashing inflatable member with latex or other sealing agent.
Lifeboat not folded as specified.
Topping-off valves not installed as specified.
Inflation cylinders not fully charged to the weight and pressure specified.
Banding straps not installed as specified.
Painter rope missing, not attached, not as specified.
Boat not balanced as specified.
Pull force on painter line to activate the inflation system exceeds 40 pounds.
Sea painter line not extending under the bottom of the sealing band.
Hand pump and adapters not operable, not stowed as specified.
Boarding net damaged, not as specified.
Canopy damaged, not as specified.
Sea anchor "dee" ring not installed, not as specified.
Sea anchors missing, not installed or stowed as specified.
Cylinder holders not installed as specified.
Weight of the completed, packed lifeboat, ready for stowage exceeds maximum weight specified.
Materials used other than those specified.
Survival gear missing or not as specified.
Coated cloth of other than inflatable compartments damaged, not properly cured, not as specified.
Hull and gunwale tubes not joined as specified.
Ends of cross tubes not as specified.
Equipment incomplete, components not as specified, not stowed as specified.
Various "dee" rings will not hinge, material not as specified, not installed as specified.
Snap and lift fasteners damaged, not installed, not as specified.
Length and width of the finished lifeboat not within the tolerances specified.
Hull, gunwale, cross tube, diameters not within the tolerances specified.
Unauthorized (in the opinion of the Navy inspector) repairs or rework.

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TABLE II. Types of defects - Continued.

Defects
Oars damaged or missing, not as specified, not stowed as specified.
Lights not installed as specified, batteries missing.
Canopy closure fasteners not installed, not as specified.
Righting lines missing, not installed as specified, not as specified.
Radar reflective screen missing, not installed or stowed as specified, not as specified.
Metal parts not coated as specified.
Floatable knife missing, not as specified.
Inner canopy vent holes not as specified.
Identification marking not legible, not as specified.
Packing of survival gear not as specified.
Instruction manuals missing, not as specified.
Caution tag or instruction lettering missing, not as specified.
Rope of line ends not fused or seized as specified.
Rain catcher tube construction not as specified, installation not as specified.
Inflatable floor not as specified.
Sealing clamps missing or not as specified.
Rescue line not as specified, not installed as specified.
Survival gear bag attachment lines missing, not installed as specified.
Sewing thread not as specified.
Life line not as specified, not installed as specified.
Closed container not sealed as specified.
Fasteners for stowage packets not operable.

4.7 Test conditions. Unless otherwise specified herein, all tests on coated cloth required by this specification shall be conducted under the standard atmospheric conditions specified in FED-STD-191. Samples shall be conditioned in the standard atmosphere for not less than 24 hours before testing. The tests required by 4.8 shall be conducted at an atmospheric pressure of 27 to 32 inches of mercury (Hg) and a temperature as specified by the applicable test of 4.8.

4.7.1 Temperature correction. If the ambient temperature changes during the air pressure tests, correction shall be made by calculating the variation in pressure due to temperature change as follows:

(a) Subtract the final temperature from the initial temperature (a rise in T is a minus, and a fall is a plus).
(b) Multiply the difference by 0.031 lb/in^2 for every change of 1°F.
(c) Add this correction to the recorded pressure.

4.7.2 Barometric pressure correction. If the barometric pressure changes during the air pressure tests, correction shall be made by calculating the variation in pressure due to barometric pressure change as follows:

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(a) Subtract the final barometric pressure from the initial (a rise in P is a plus, and a fall is a minus).
(b) Multiply the difference by 0.049 lb/in^2 for every 0.1 inch Hg change.
(c) Add this correction to the recorded pressure.

4.8 Tests. Failure of inflation system to operate properly, or painter line pull force greater than 40 pounds shall be cause for rejection.

4.8.1 Inflation. The lifeboat, complete in its container, shall be inflated in accordance with the instructions located on the end of the container to ascertain that the case will open properly and the inflation system performs properly. (The boat shall inflate to its design shape in not greater than 30 seconds.) The lifeboat shall be allowed to stand for 1 hour, minimum, or until the relief valves have closed, to allow the air in the tubes to come to room temperature. Pressure readings shall be taken through the valves of each compartment of both upper and lower flotation tubes. (Pressure in any compartment shall be not less than 2.3 lb/in^2 nor greater than 3.8 lb/in^2 when tested under the conditions specified in 4.7.) Failure of the case to open properly or failure of inflation system to operate properly shall be cause for rejection. Necessary precautions shall be taken to ensure that only fully charged, tested cylinder assemblies for proper inflation at sea level are shipped with the lifeboats. Utilization of a full size dummy survival gear bag is permitted during initial performance of this test.

4.8.2 Overpressure leakage. Upon completion of the inflation test as specified in 4.8.1, the relief valves shall be blanked, and the pressure in the lifeboat shall be increased to 5 lb/in^2 and allowed to stand for not less than 10 minutes. The pressure shall then be checked and readjusted, if necessary, to 5 lb/in^2. If the pressure at the end of an additional 20-minute period (minimum) is less than 4.6 lb/in^2, the lifeboat shall be rejected. Care shall be exercised not to introduce a differential pressure across the bulkheads during this test.

4.8.2.1 Tubes. One compartment of the upper flotation tube and one compartment of the lower flotation tube shall be inflated by air to 3.8 lb/in^2 through the topping-off valves and allowed to stand for not less than 10 minutes. The pressure shall then be checked and readjusted, if necessary, to 3.8 lb/in^2. If the pressure at the end of an additional 20-minute period (minimum) is less than 3.3 lb/in^2, the lifeboat shall be rejected. The two compartments under test shall then be deflated and the same test conducted on the remaining two compartments.

4.8.2.2 Floor. The inflatable floor shall be subjected to the test specified in 4.8.2. If the final corrected pressure in the inflatable floor is less than 4.6 lb/in^2, the floor shall be rejected. Upon completion of this test, the lifeboat shall immediately be subjected to the test specified in 4.8.3.

4.8.3 Working pressure and bulkhead. One compartment of the lower flotation tube and one compartment of the upper flotation tube shall be inflated by air to 2.0 lb/in^2 through the topping-off valves and allowed to stand for 1 hour. The pressure shall then be checked and readjusted as necessary to the original pressure of 2.0 lb/in^2. After standing 6 hours, the pressure shall be measured. The two compartments under test shall be deflated and the same test conducted on the remaining two compartments. If any compartment has a corrected pressure of

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less than 1.7 lb/in^2 after 6 hours, the life boat shall be rejected. If during the 24 hour test period there is a temperature variation greater than 10°F, the test is invalid and shall be repeated. The entire lifeboat and inflatable floor shall then be inflated by air to 2.0 lb/in^2 and pressure readjusted as above after 1 hour. After standing 24 hours, the pressure shall be measured. If the inflatable floor has a corrected pressure of less than 1.5 lb/in^2 after 24 hours, it shall be rejected. Depending upon the total number of repairs to the inflatable portions of the boat (excluding the inflatable floor), if any compartment has a corrected pressure after 24 hours or less than the minimum allowable pressures shown below, the lifeboat shall be rejected.

Number of repairs	Minimum allowable pressure after 24 hours (lb/in^2)
5 or less	1.5
6 - 10	1.55
11 - 15	1.6

4.8.4 Life line patch. Life line patches shall be tested for tensile strength in a fixture. The direction of the pull shall be perpendicular to the patch. This test is normally required only on lifeboats submitted for first article inspection, but should be applied at any time that questionable quality is evident (see 3.5.19).

4.8.5 Lap seam strength. Each lifeboat lap seam shall be certified by subjecting the test samples (fabricated in accordance with 4.5.1) to the tests of 4.8.5.1 and 4.8.5.2. In the event of lap seam tests failure, the lifeboat shall be subjected to the testing of 4.8.5.3.

4.8.5.1 Tensile strength. One untaped seam sample for each lifeboat shall be tested for tensile strength in a tensile machine. The seam strength shall be sufficient to withstand a pull test load greater than the strength of the coated fabric.

4.8.5.2 Vulcanization. The second untaped seam sample shall be tested for proper vulcanization. When subjected to a tensile load of 40 pounds per inch of width, at a temperature of 200 degrees F + 5°F, the untaped seam shall show no signs of slippage after 24 hours. Slippage has occurred if the unit change in seam overlap is greater than the unit stretch of the adjacent fabric.

4.8.5.3 Failure. In the event there is a failure of either seam sample to pass the tests of 4.8.5.1 and 4.8.5.2, the lifeboat represented by the individual specimens which fail the test shall be further tested by cutting out seam samples from the boat itself (two seam samples from each boat). The samples shall be cut from both top and bottom air holding tubes. All load carrying support provided by the seam tape shall be removed. Only the 24-hour deadload test needs to be completed. If the boat seams pass this test, the boat is acceptable, subject to repair and subsequent successful completion of the overpressure leakage test of 4.8.2. Lifeboats whose seam samples fail this test shall be rejected.

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4.8.6 Low temperature inflation. The lifeboat with all equipment and its container shall be conditioned in an environmental chamber as follows. The lifeboat shall be conditioned until the temperature is reduced to minus 22 ± 2°F and then the lifeboat shall continue to be cold-soaked for not less than 12 hours at this temperature. Thermocouples shall be installed on each inflation cylinder and in the interior of the folded boat to verify that the desired temperatures have been reached. The container may be closed or open. If conditioning is done with the container open, the container shall be closed and banded as specified in MIL-C-24491 prior to inflation of the lifeboat. The lifeboat shall be inflated by means of the normal actuating assembly. The lifeboat shall fully inflate to design shape with all canopy bows up in not more than 30 seconds. Each flotation tube compartment shall have an internal pressure of not less than 0.5 lb/in^2 at that time. Pressure readings shall be taken at the topping-off valve for each compartment.

4.8.7 High temperature inflation. The lifeboat with all equipment and its container shall be conditioned in an environmental chamber as follows. The lifeboat shall be conditioned until the temperature is elevated to 160 ± 5°F and then the lifeboat shall continue to be heat-soaked for a minimum of 12 hours at this temperature. Thermocouples shall be used to verify temperature as specified in 4.8.6, and the lifeboat shall be packed and inflated in the manner specified in 4.8.6, except that the ambient temperature shall be 160 ± 5°F. The lifeboat shall be fully inflated in not more than 30 seconds. All relief valves shall operate as specified in MIL-I-24490. At no time shall the pressure in any flotation compartment exceed 5.0 lb/in^2.

4.8.8 Drop. Lifeboat with all equipment, and packed in its container, shall be drop tested as specified in MIL-C-24491.

4.9 Inspection of packaging. Packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation. Protective preservation and preparation of equipment to be furnished with each lifeboat shall be in accordance with 3.5.24, 3.5.25, and 3.6 herein.

5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2).

5.2.1 Levels A, B and C. Each stowage container including the lifeboat shall be packed in an open wood crate conforming to MIL-C-52950, style and class selection at the contractor's option. Anchoring, closure, and strapping shall be in accordance with the appendix to the crate specification.

5.2.2 Commercial. Each stowage container including the lifeboats shall be packed in accordance with ASTM D 3951. Method of packing shall facilitate handling and storage by fork lift trucks.

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5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packs and shipping containers shall be marked, including bar coding, in accordance with MIL-STD-129. The lifeboat, and any other included item with shelf life requirements, shall include the date of manufacture (month and year). The fiberglass container and the shipping container shall be marked with the month and year of shipment and the boat serial number.

5.3.1 Hazardous material marking. Interior packages and items, including each lifeboat container and exterior shipping container, shall be marked and labelled to identify the compressed air, signal device kit, or any other hazardous material contained within the packed lifeboat. Markings and labels conform to DOT Code of Federal Regulations, Title 49, 171-178.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lifeboat described herein is intended for shipboard installation to be used in the event the ship must be abandoned at sea.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article inspection is required (see 3.1).
(d) Level of packing required (see 5.2).
(e) Special marking required (see 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3, 4.4, 4.5.1, 4.5.2, and appendix A	DI-MISC-80678	Certification/data report.	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (see 6.3) production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging or preparation of delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.5.23.1, 3.5.23.11.1, and 3.10.

6.7 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.8 Repairs. The number, type, and location of repairs for each boat shall be recorded and forwarded with those boats submitted for first article inspection (see 4.4).

6.9 Government representative. While the contractor performs the test in 4.8.5 and records the results, the Government representative will witness the test and take possession of the test seams identified as to the lot of boats they represent. These samples and the test results will be held for a minimum of 2 years.

6.9.1 Only fully tested inflation gear be shipped with the lifeboat shall be verified by the cognizant Government representative (see 4.5.1).

6.10 Subject term (key word) listing.

Abandon-ship equipment
Inflation system
Rigid stowage container
Survival gear

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 1940-N085)

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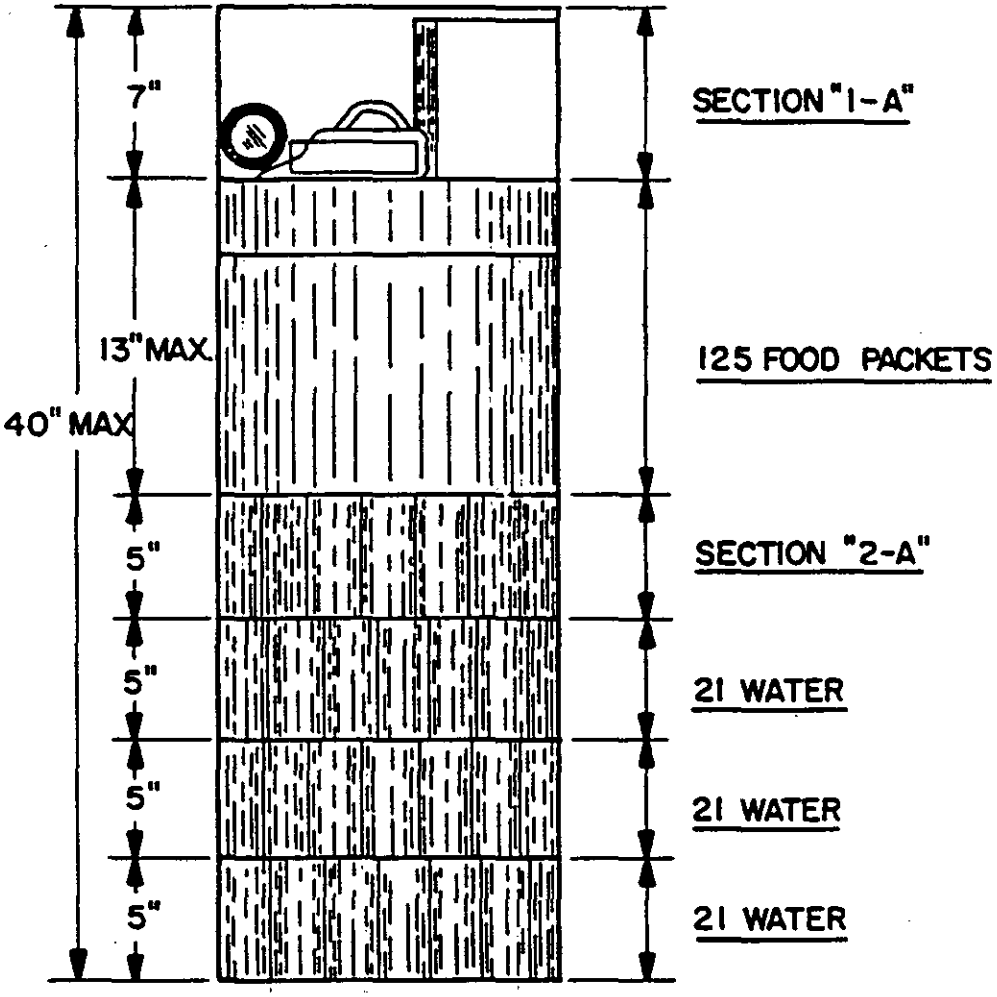


FIGURE 1. Packing of survival gear bag.

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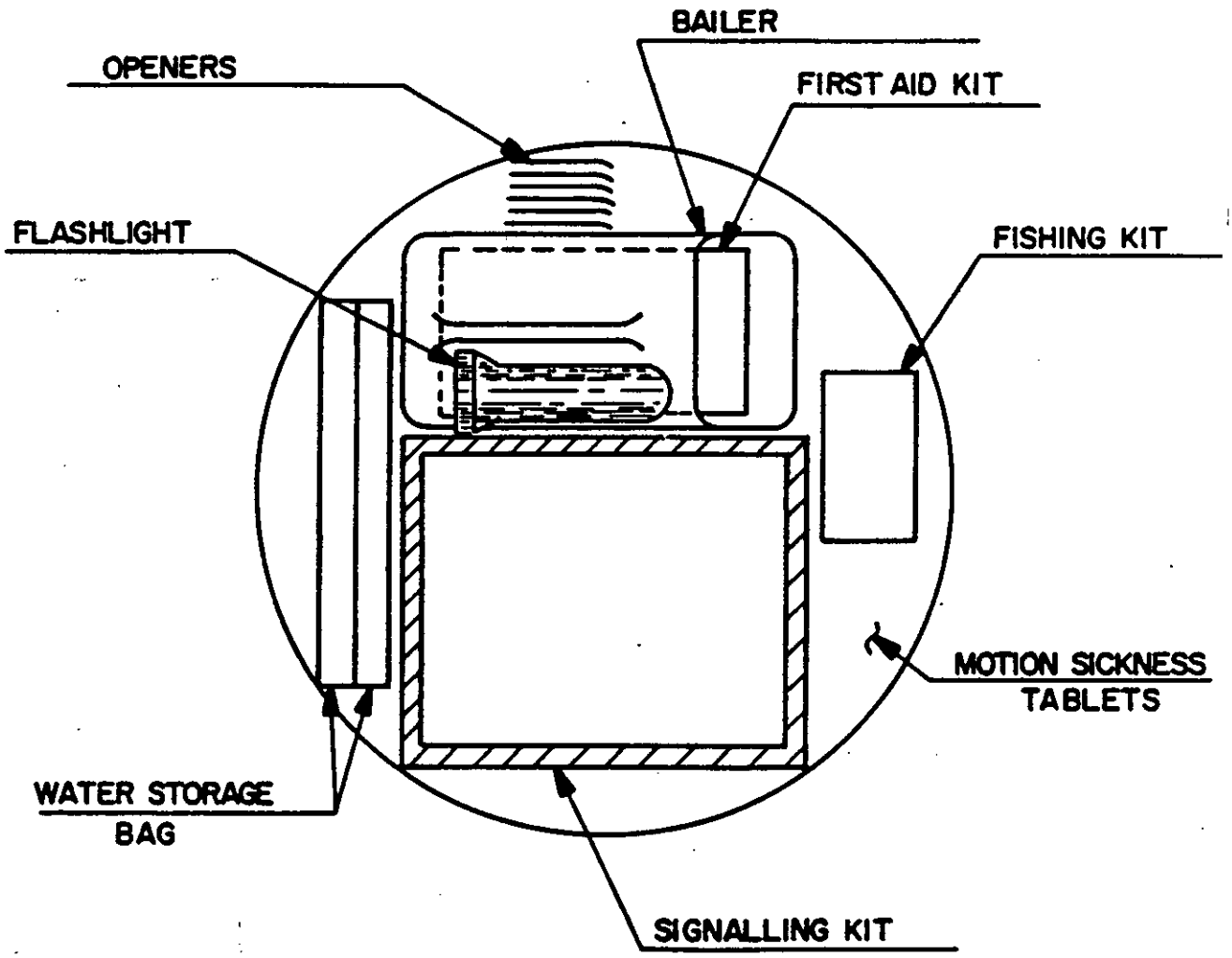
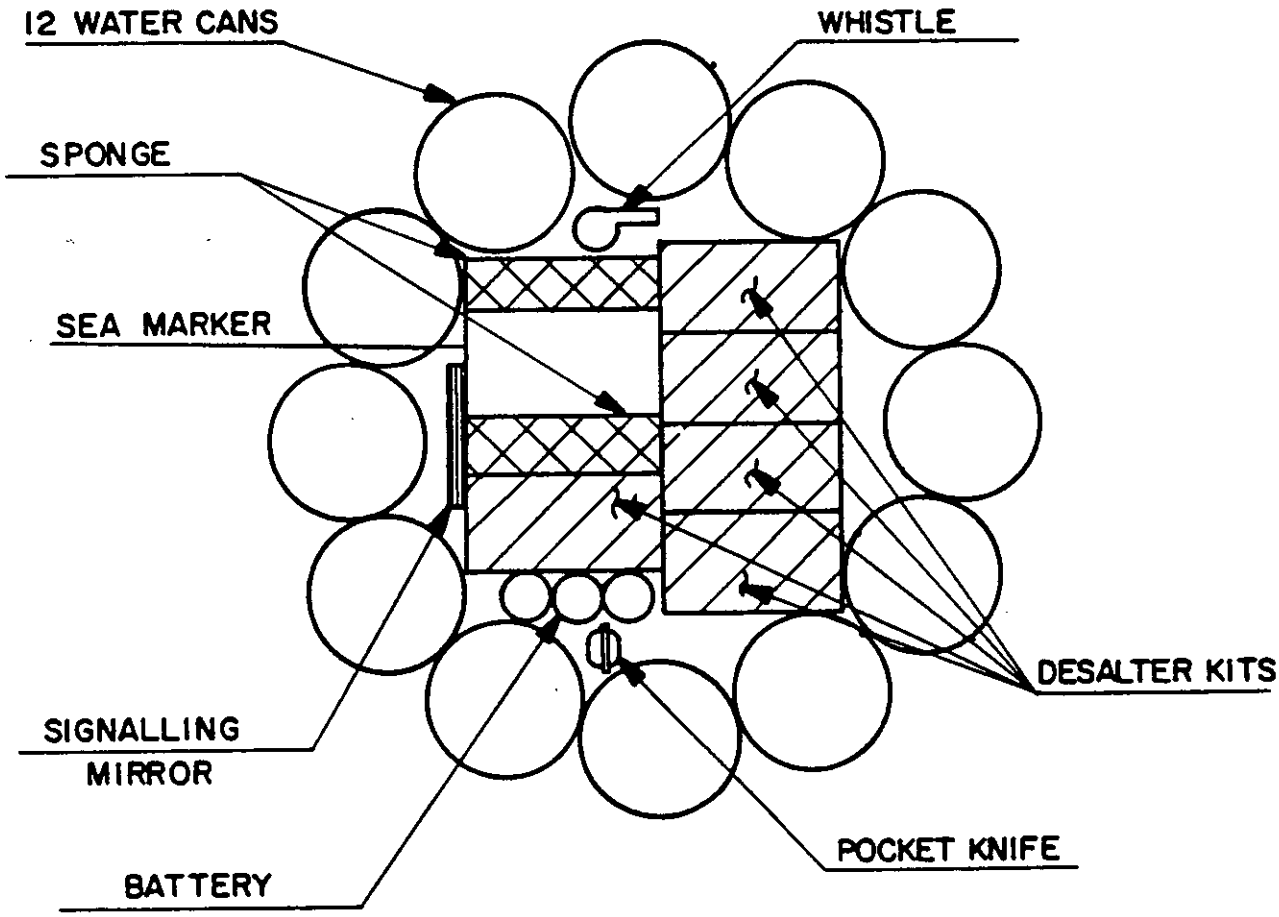


FIGURE 1. Packing of survival gear bag (section 1-A) - Continued.

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SECTION "2-A"

FIGURE 1. Packing of survival gear bag (section 2-A) - Continued.

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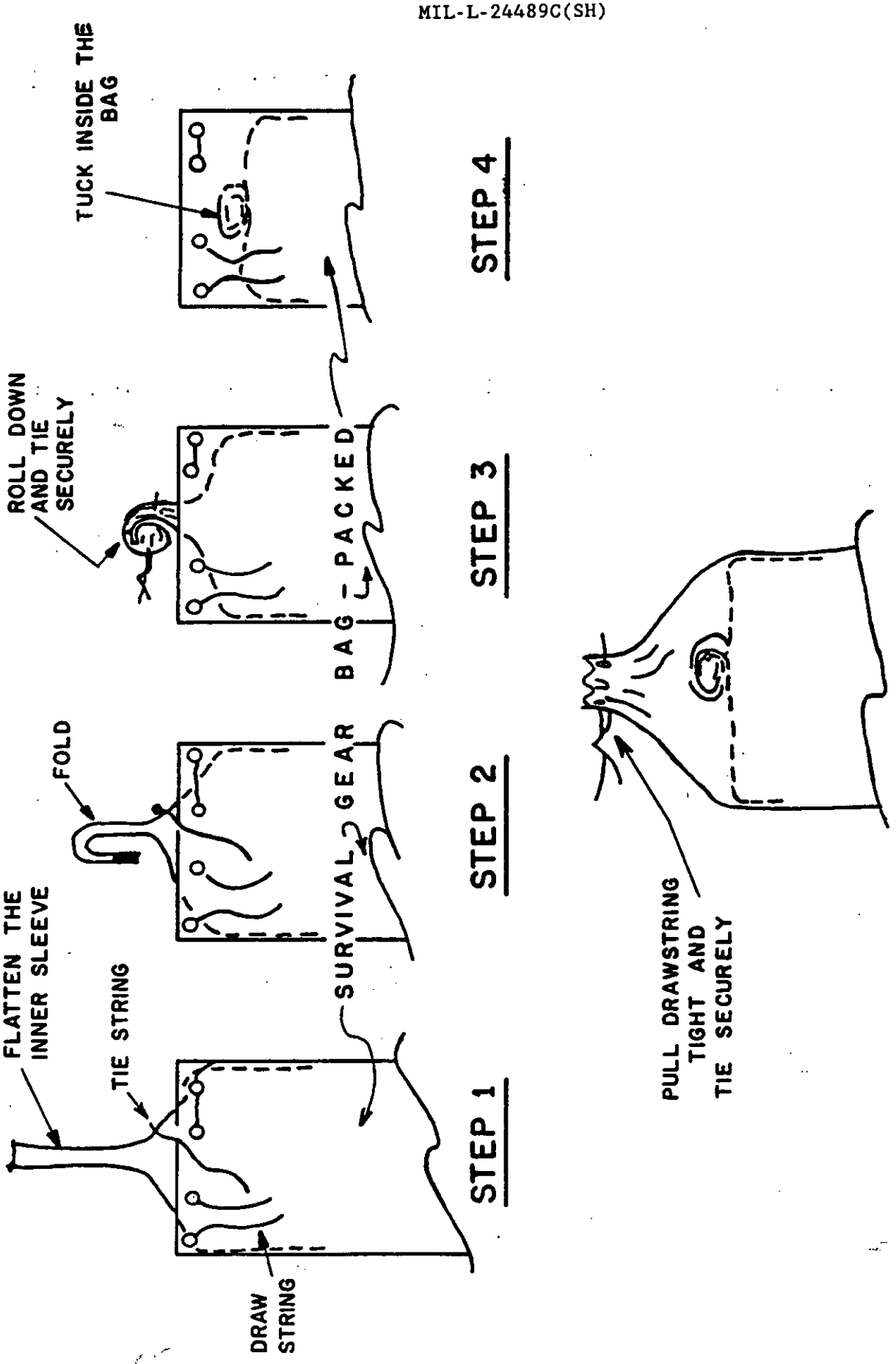


FIGURE 2. Closing the survival gear bag.

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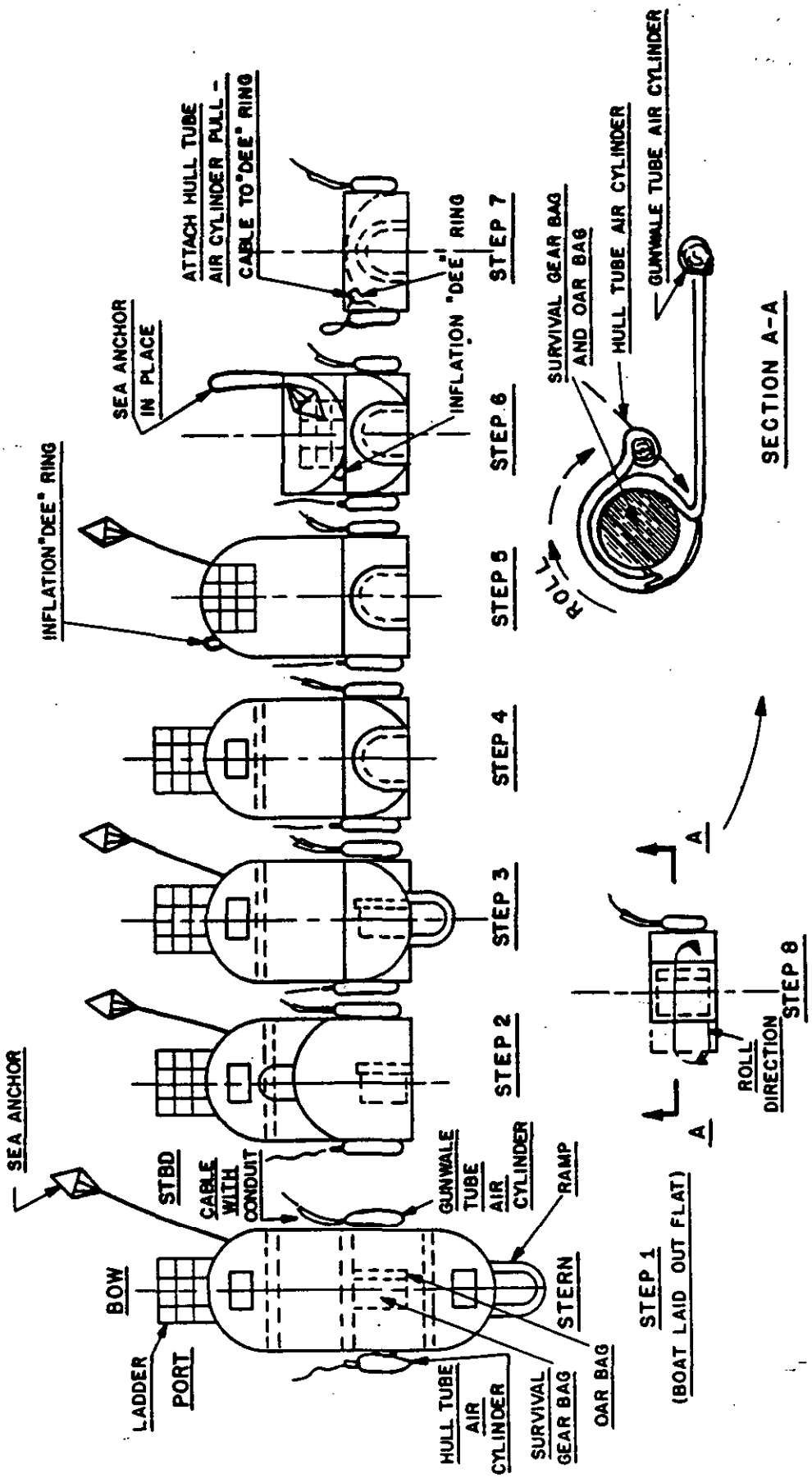


FIGURE 3. Folding instructions.

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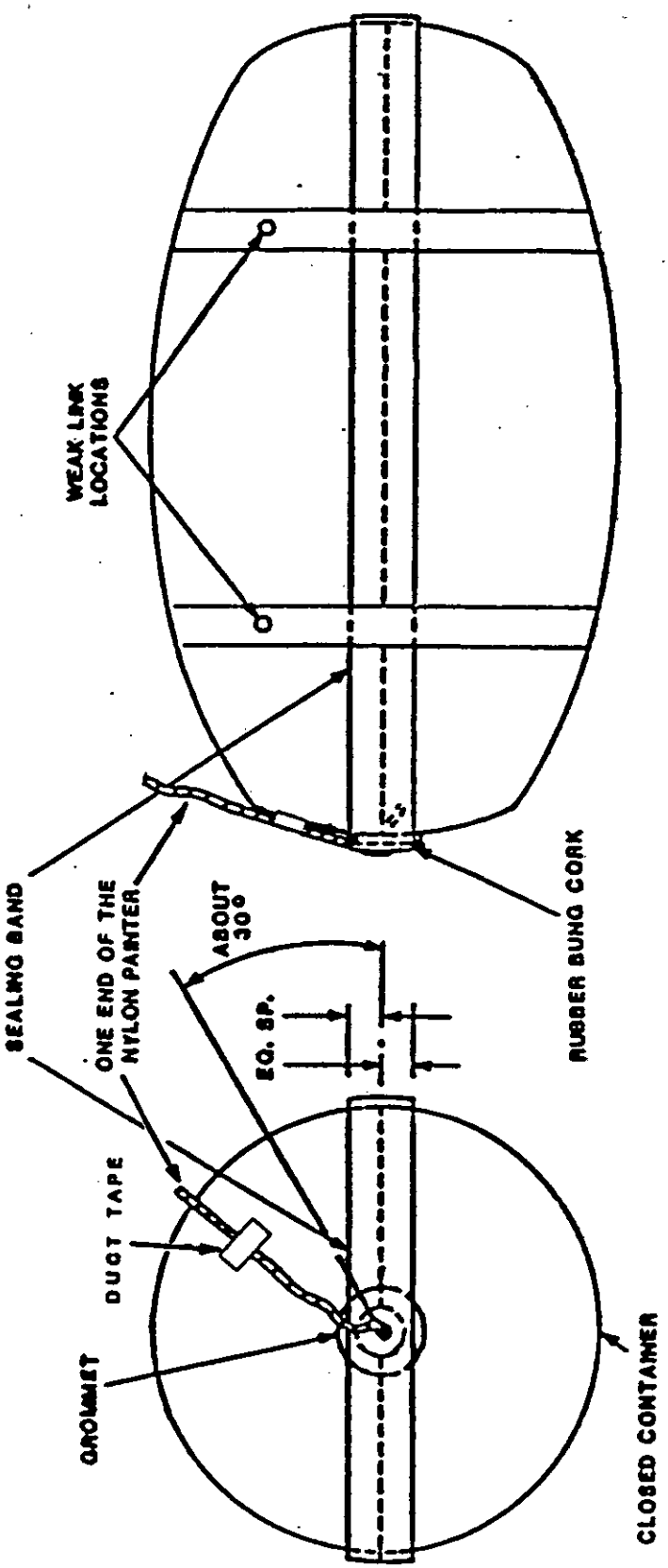


FIGURE 4. Lifeboat container sealing and securing arrangement.

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APPENDIX A

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the certification/data report when specified in the contract or order. This appendix is mandatory only when data item description DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION CONTENT

30.1 Certification of first article inspection. A certification/data report shall provide the results of the detailed inspection and tests of the lifeboat (4.3 and 4.4):

(a) Certification each type of rope or line used (see 3.2.1) was prepared in accordance with appropriate specification (see 4.4).
(b) Record of number, type, and location of repairs for lifeboat complete with equipment and stowage container.

30.2 Quality conformance inspection. A certification/data report shall provide results of all inspections and tests specified as specified below (see 4.5):

(a) Inspections were conducted by contractor under the direct surveillance of the Government Representative (see 4.5).
(b) Verification that only fully tested inflation gear be shipped with lifeboat (see 4.5.2).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-24489C(SH)	15 April 1994

3. DOCUMENT TITLE

LIFEBOAT, INFLATABLE, MARK 6 25-PERSON CAPACITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Mr. Richard Branch, Code 1271.3 Naval Surface Warfare Center Norfolk	804-444-9164	DSN 564-9164
c. ADDRESS (Include Zip Code) Commander Sea 93R42 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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